

Unit: 3 Data

Short Questions

1. What are digital symbols, and why are they important in computing?

⇒ Digital symbols are the binary digits 0 and 1 used by the computer. Digital symbols are important because computer only understand data in 0's and 1's [i.e. if there is incoming voltage or not.]

2. Differentiate between Online and Offline peripherals.

⇒ Online peripherals are those peripherals that are directly connected to the system whereas Offline peripherals are those peripherals that are not connected in real time.

3. What is the purpose of coding schemes like ASCII and unicode?

⇒ Coding schemes like ASCII and Unicode provide a standard representation of characters as numeric codes

enabling computers to store, process and exchange text data among different systems and languages.

Explain
4) ~~Discuss~~ the difference between verification and validation.

⇒ Verification ensures data is copied/entered correctly whereas validation ensures data is logical and complete.

5) What are the main stages of data input?

⇒ The main stages of data input are:

- Data Collection
- Data Preparation
- Data Entry
- Verification and Validation.
- Input Methods.

Long Question:

Write a detailed note on the representation of data and the stages of data input.

⇒ Data representation and input are fundamental to processing & storage. Storing data in computer.

Data in computer is stored in binary form of 0 and 1. It is so because computers operate on the digital logic of ON and OFF.

Different types of data are stored/represented as given below.

(i) Numeric data: As signed/unsigned integers and floating-point representation.

(ii) Character data: Stored using coding schemes like ASCII where each character is assigned a binary code.

(iii) Image data: Stored as pixels with colored data.

(iv) Audio data: Stored as sampled waveform

(v) Video data: Stored as sequence of image frames with audio.

(vi) Boolean: Stored as True (1) or False (0).

Data input is carried out in multiple steps:

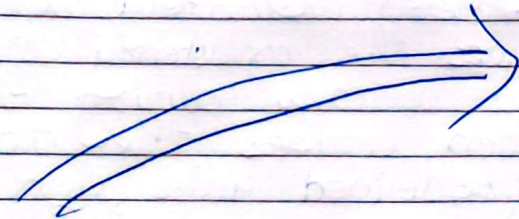
(i) Data collection: Collecting raw data from the source.

(ii) Data preparation: Editing coding and formatting data.

(iii) Data Entry: Entering data into the computer system.

(iv) Verification: Ensure the data is entered correctly.

(v) Validation: Ensure data is entered properly.



2. Explain the role of Online and Offline peripherals

⇒

Feature	Online Peripherals	Offline Peripherals
Connectivity	Always Connected	Connected as needed
Operation	Real-time interaction	Independent operation
Samples	Keyboard, Webcam	Printer, Scanner
Advantage	Immediate functionality	Portability & flexibility
Disadvantage	Requires constant link	Delayed interaction

3. Discuss the importance of Verification and Validation in data processing.

⇒ Verification refers to the process of checking / ensuring whether the data is transferred accurately between systems. It ensures the consistency of data as well as accuracy during transfer. Validation ensures the data meets the predefined rules and constraints before being processed. It checks for logical consistency and adherence to standards.

Importance of Verification and Validation.

- Ensures compliance with regulations and standards.
- Enhances data reliability and accuracy.
- Reduces operational risks and decision-making errors.
- Prevents errors in data processing.

4. Compare the functionality and applications of ASCII and Unicode coding schemes.

Feature	ASCII	Unicode
Full-Form	American Standard Code for Information Interchange	Universal Character Encoding
Bit length	7 or 8 bits	8 to 32 bits
Supported Characters	128 (basic)	Over 140,000
Language Support	English only	Multiple languages
Usage	Older Systems	Modern Systems & Applications

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5. Describe the input methods and media commonly used in computer systems

⇒ Input Methods:

- (i) Manual input - keyboard, mouse.
- (ii) Automatic input - Barcode readers, QR
- (iii) Direct input - Touch screen, biometric
- (iv) Audio input - Microphones with speech recognition.

Input Media:

- (i) Paper documents (forms, questionnaire)
- (ii) Magnetic media (hard disk, tapes)
- (iii) Optical media (CDs, DVDs)
- (iv) Solid-state media (SSDs, USBs)